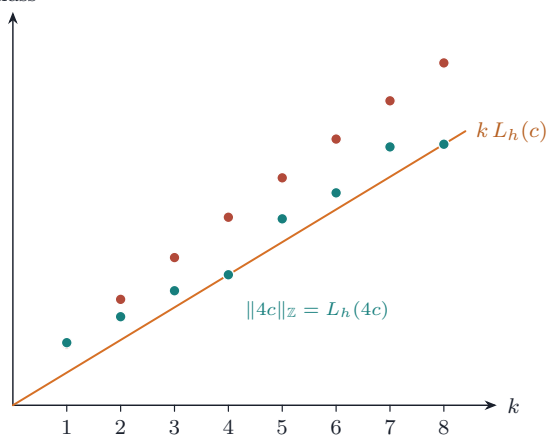


mass



$L_h(kc) = k L_h(c)$: the minimum over real

currents, attained by a smooth strongly positive form

$\|kc\|_{\mathbb{Z}}$ for a class that is not algebraic:

- above the line for every k , and $\|kc\|_{\mathbb{Z}}/k \rightarrow L_h(c)$ (Federer)

an algebraic class: the gap closes at

- some k and at its multiples, where the minimiser is an effective cycle

the heights of the dots are schematic;
the line and the limit are theorems

The Mumford square $Y = X \times X$, $\theta_Y = \theta \otimes 1 + 1 \otimes \theta$

$$\theta_Y^6 = \sum_k \binom{6}{k} \theta^k \otimes \theta^{6-k}$$

k	0	1	2	3	4	5	6
$\binom{6}{k}$	1	6	15	20	15	6	1
$\int_Y \pi_0 \theta^k \otimes \theta^{6-k}$	0	0	0	144	0	0	0
$\int_Y \pi_{ij} \theta^k \otimes \theta^{6-k}$	0	0	0	0	0	0	0

$$\int_Y \pi_0 \theta_Y^6 = 20 \cdot 144 = 2880, \text{ so}$$

$$L_{\theta_Y}(\omega_a) = 2880 a_0 / 6! = 4a_0$$

$L_{\theta_Y}(\omega) = 0$ for an exceptional class with $a_0 = 0$

$$L_{\theta_Y}(\theta_Y^2) = 8!/6! = 56; 4\theta_Y^2 = [D \cap D']$$

for $D, D' \in |2\theta_Y|$, of mass 224

open: an integral current of mass $56NM$ in the class dual to $N(M\theta_Y^2 + \omega)$; known at every CM point with N, M depending on the point; the same N, M at infinitely many CM points would give every member